wavetopo — companion notes: variants, periodic media, a modal bracket, and valley-Hall guiding

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Status of these notes. These sections were split out of the main manuscript (`dolfinx_wave_control.tex`), whose question is how much control fiber orientation alone exerts on in-plane elastic waves, by what mechanism, and within what limits. Each section below answers a *different* question and is preserved verbatim from the earlier draft, with its own caveats. Three caveats from the main paper apply here. (i) Single-frequency focus-energy gains of finite plates include resonant field build-up and must be decomposed into redirection and amplification before they are compared. (ii) Single-frequency cloak reductions are narrowband. (iii) The periodic band-gap claims are subject to the dense-$\mathbf{k}$ caveats stated in the text.

0.1 Multi-target beam shaping (two foci and a null)

The same machinery handles a genuinely multi-objective goal by a signed region weight. We ask the plate to *maximize* energy in two separated focus regions while *minimizing* it in a third between them: setting $w = +1$ on the two targets and $w = -\lambda$ on the null ($\lambda = 1$) and maximizing $J = \int_\Omega w|\mathbf{u}|^2$ drives a fiber beam splitter. The straight-fiber baseline illuminates the three regions almost equally (target/null contrast 0.0); the optimized orientation splits the wave into the two off-axis foci and holds the centre as a local dark node, raising the target/null contrast to 18 while the foci-to-null energy swing against the baseline reaches 1500$\times$ (Figure 1). (The whole field amplifies off the very dim straight-fiber baseline, so the meaningful measure of the null is its *contrast* against the neighbouring foci, not its absolute level.) Only the signed objective changed — the physics, adjoint, parametrization and toolpath pipeline are identical to the single-focus lens.

1 Designing around prescribed through-holes

The reference work ties the curvature limit directly to geometry: a *prescribed circular void* (a passive, zero-density through-hole) of radius R imposes a local minimum turning radius on the fibers wrapping it, so the manufacturable curvature limit is $\zeta_{\text{all}} = 1/R$. We reproduce this rule in both wave-control settings.

The holes enter the physics *geometrically*: each disk is cut from the domain with `gmsh`/OCC and the triangulation conforms to its boundary (§**??**, `wavetopo/dolfinx_mesh.py`), so the rim is exactly circular and traction-free through the natural boundary condition of the weak form, with the element size halved on the rim where the diffraction happens. This matters more than it may appear. The obvious alternative — representing the hole as a SIMP soft void ($z = 0$, so $z^p \mathbf{D}_f \to 0$) on a structured grid — replaces the circle by the set of elements whose centroid lies inside it. The resulting boundary is a staircase that scatters off its own corners, concentrates spurious energy

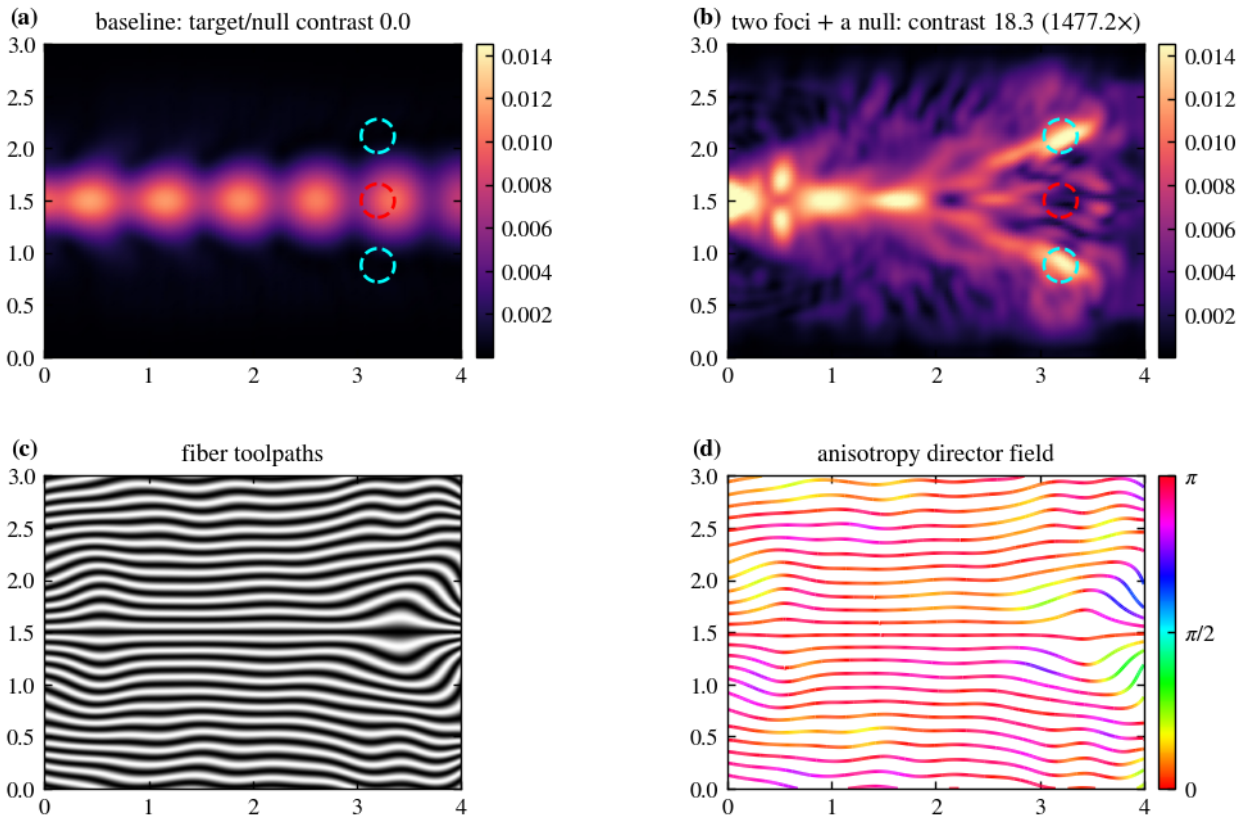


Figure 1: Multi-target beam shaping. The objective maximizes energy in the two targets (cyan) and minimizes it in the null (red). Top: straight-fiber baseline vs. the optimized two-focus-plus-null design (target/null contrast 0.0 $\rightarrow$ 18). Bottom: the phase-field fiber toolpaths (a beam splitter) and the convergence of J. Same solver, adjoint and parametrization as the single-focus lens; only the signed region weight differs.

on the rim, and defines the radius only to within one element: precisely the radius R that the curvature rule $\zeta_{\text{all}} = 1/R$ is tied to. Conforming to the geometry is the substantive reason these problems are solved in `dolfinx` rather than in the structured Q4 code used for the periodic cell of §2, where (the geometry being the density field itself) there is no boundary to conform to and a structured grid is the better tool.

1.1 Wave lens with two circular through-holes

We cut two symmetric circular through-holes ($R = 0.28$, so $1/R \approx 3.57$) into the beam path of the lens and again maximize the focus energy. Without a curvature limit the optimizer routes the fibers through the gap between the holes and around their flanks, refocusing the diffracted field with a 49× gain but at the cost of tight bends where the tows crowd past the voids ($|\zeta|_{\max} = 5.7$). Imposing the through-hole limit $|\zeta| \leq 1/R$ (Eq. (**??**)) relaxes those bends to the radius of the holes themselves: the peak curvature falls to 3.55, just inside the limit $1/R = 3.57$, and the gain settles at 17× (Figure 2). The toolpaths now wrap the prescribed voids with arcs no tighter than the holes — a directly printable layout. Note that the constrained design is *feasible*, not merely penalized: the reported $|\zeta|_{\max}$ lies below ζ_{all} rather than near it from above.

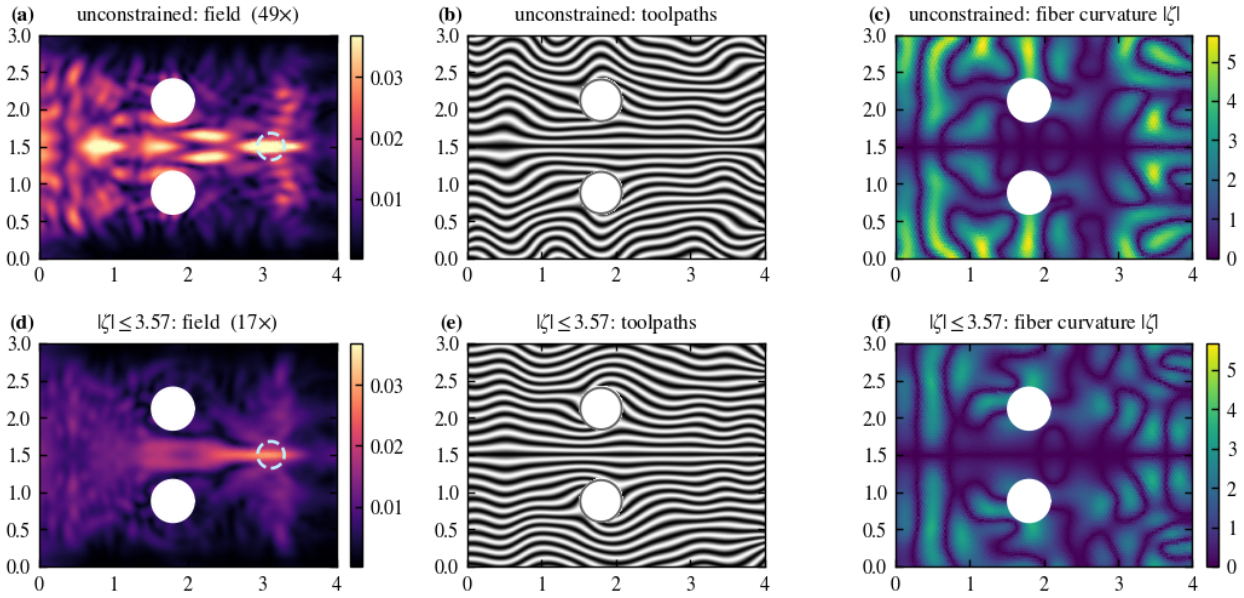


Figure 2: Wave lens with two prescribed circular through-holes (white). Top row: unconstrained — the fibers route around the voids and refocus the wave, with tight bends ($|\zeta|_{\max} = 5.7$). Bottom row: with the through-hole limit $|\zeta| \leq 1/R \approx 3.57$, the toolpaths wrap the holes with arcs no tighter than the holes ($|\zeta|_{\max} = 3.55$) at a modest focus cost. Columns: field, toolpaths, fiber-curvature $|\zeta|$ map. The dashed circle in the field panels marks the focus target the lens is optimized to fill.

1.2 Asymmetric through-holes: the design re-optimizes per geometry

The symmetric layout of Figure 2 could give the misleading impression that the fiber field is fixed and only the holes move. It is not: the orientation is re-optimized from scratch for each hole geometry. To make this explicit we place two *asymmetric* through-holes — different centres and radii, breaking the $y = L_y/2$ mirror symmetry — and drop the symmetry constraint on the design entirely (a plain, non-symmetric CS-RBF support grid). The optimizer returns a visibly *asymmetric* fiber field that threads the wave through the off-centre gap between the voids and re-focuses on the target, a 50× gain with peak curvature $|\zeta|_{\max} = 4.24$ (Figure 3). Here the curvature limit is set by the *smaller* of the two holes, $\zeta_{\text{all}} = 1/0.24 = 4.17$, and we report honestly that the converged design ends marginally *above* it (4.24 vs. 4.17, a 1.7% overshoot): the constraint is imposed by the smooth quadratic penalty of §**??** under continuation, which drives the curvature down to the neighbourhood of the bound but does not enforce it exactly. The symmetric case of §1.1 does land strictly inside its bound (3.55 vs. 3.57); this one does not quite, and a hard constraint (or a larger terminal μ) would be required to certify feasibility. The toolpaths and the director map wrap the two holes differently, exactly as the differing local geometry demands — the design is genuinely solved anew, not mirrored.

1.3 Routing energy around a joint: a curved delivery corridor

The through-hole studies so far treat the void as an obstacle to be worked around. A bolt hole or a fastened joint is more than that: it is the place where a panel *fails*. Fatigue cracks nucleate at the rim, so vibrational energy concentrating there is precisely what a designer wants to avoid.

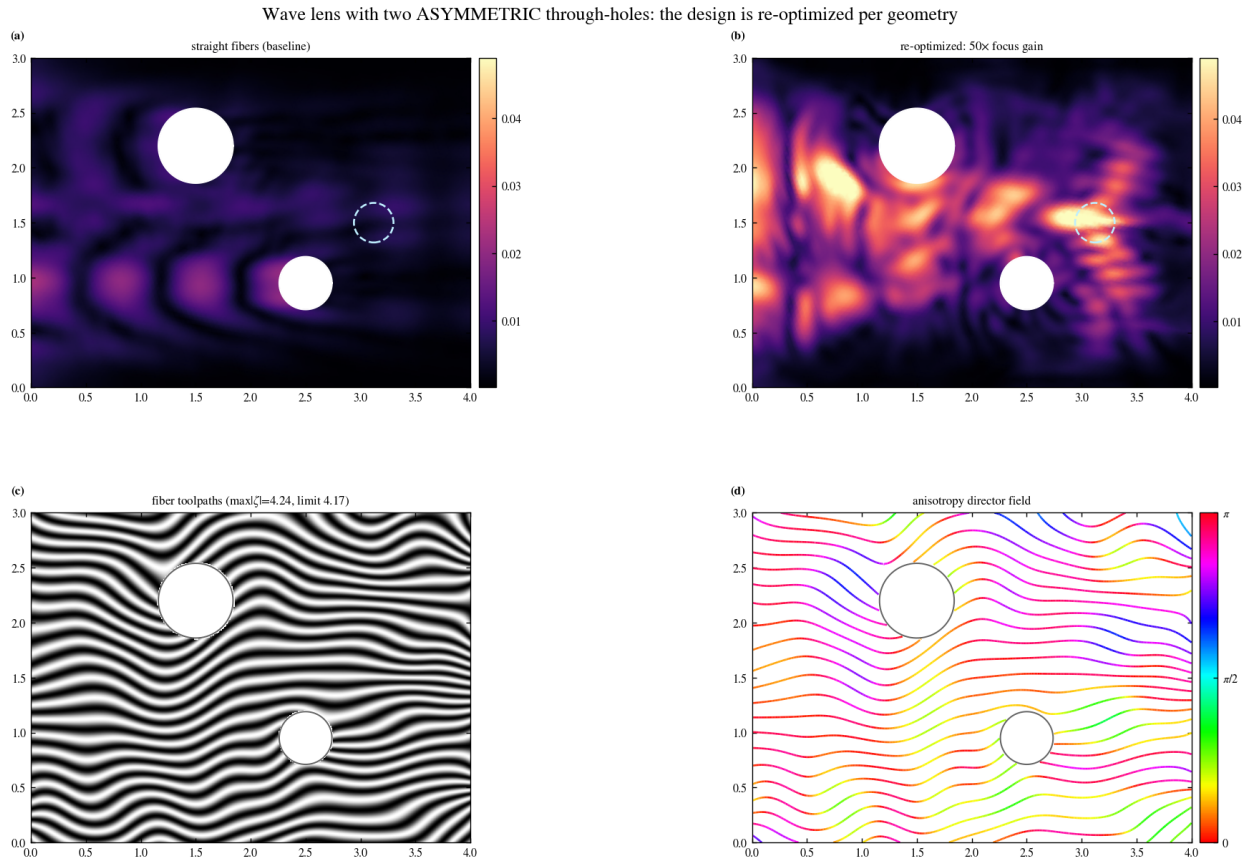


Figure 3: Wave lens with two *asymmetric* through-holes (no symmetry imposed). Top: straight-fiber baseline and the re-optimized field (50× focus gain). Bottom: the phase-field fiber toolpaths and the anisotropy director map — both asymmetric, wrapping the differently placed/sized voids as their local geometry requires. This is a genuine per-geometry re-optimization, not a mirrored copy of Figure 2. The dashed circle in the field panels marks the focus target.

This suggests a design problem the reference work does not pose — route the energy along a prescribed *curved* path that detours around the joint, delivering it downstream while keeping the fatigue-critical rim quiet.

We place the joint (radius $R_J = 0.40$, cut conformally from the mesh) squarely on the straight line between the drive and a delivery port, so the direct path is blocked and the energy *must* detour. Three regions define the goal: a curved corridor whose centreline is a smooth bump lifting the path over the joint,

$$y_c(x) = y_J + A\, e^{-((x-x_J)/s)^2}, \qquad \Omega_{\mathrm{cor}} = \left\{\, |y - y_c(x)| < h \,\right\}, \tag{1}$$

an exit port Ω_{exit} downstream of the joint, and a guard annulus $\Omega_{\mathrm{gd}} = \{R_J \le r < R_J + g\}$ hugging the rim. Both objectives are carried by a *single signed* region weight, exactly the device used for the multi-target lens,

$$w(\mathbf{x}) = \underbrace{\mathbb{1}_{\Omega_{\mathrm{cor}}\cup\Omega_{\mathrm{exit}}}(\mathbf{x})}_{\text{deliver here}} - \lambda\, \underbrace{\mathbb{1}_{\Omega_{\mathrm{gd}}}(\mathbf{x})}_{\text{protect here}}, \qquad \max_{\hat{\theta}}\ J = \int_\Omega w\,|\mathbf{u}|^2, \tag{2}$$

with $\lambda = 2.5$. Note J is genuinely *signed* — at the straight-fiber baseline the guard term dominates

and $J < 0$ — so this problem is descended with the raw gradient rather than the log-objective of §**??**, which would be undefined. The fiber curvature is limited to the joint's own radius, $\zeta_{\text{all}} = 1/R_J = 6.67$, so the tows may not bend tighter than the hole they wrap.

The straight-fiber baseline illustrates the problem: the beam launched at the aperture diffracts, spreads onto the joint, and barely reaches the port — the guard annulus carries 127× more energy than the exit, and only 22% of the downstream energy is in the corridor at all. Designing the orientation alone raises the delivered energy by 698×, cuts the guard energy to 0.17 of baseline, and lifts the downstream confinement from 22% to **97**%: the field becomes a continuous ribbon that tracks the corridor over the joint and terminates at the port (Figure 4). The design is feasible, $|\zeta|_{\max} = 6.59$ against the print-head limit 6.67.

Three modelling choices decide whether this works at all. Each was arrived at by a failure, and each is worth stating because the failures are silent — they return healthy-looking headline numbers for designs that do the wrong thing.

(i) The source must be localized. With a full-width plane-wave drive most of the field never enters the corridor, so there is nothing to channel and the result is a diffraction pattern with a bright spot at the port. A narrow aperture at the channel mouth is what makes the problem a guiding problem.

(ii) The curvature limit must be the print head's, not the hole's. Setting $\zeta_{\text{all}} = 1/R_J$ — the reference work's rule for fibers *wrapping* a prescribed void — is the wrong constraint for a guiding channel and is strictly binding: the optimizer sits pinned at the bound with the trust region collapsed and the objective still negative. The physically relevant limit is the machine's minimum turning radius, here $\zeta_{\text{all}} = 1/R_{\text{tow}} = 6.67$; the converged design uses 6.59, i.e. nearly three times the curvature the hole rule would have allowed.

(iii) The delivery reward must saturate. With a reward linear in $E_{\text{exit}}/E^0_{\text{exit}}$ the delivered energy reaches 6300× baseline, so that single term contributes 6300 against a guard penalty of 8 — a 760 : 1 imbalance — and the optimizer buys delivery by letting the joint get 20× *busier*, defeating the purpose. A logarithmic reward ($\log 6300 = 8.7$) has diminishing returns, so protecting the rim remains worth doing however bright the port already is. The gradient still costs one adjoint solve: for $J = \sum_k c_k E_k$ the adjoint weight is $\sum_k (\partial J/\partial E_k)\, w_k$, with $\partial J/\partial E_{\text{exit}} = W_E/E_{\text{exit}}$ re-evaluated at the current iterate.

Confinement is bought by the leakage penalty, and it has an optimum. Table 1 sweeps λ_{out}. Too weak and the wave diffracts around *both* sides of the joint — a physically natural but undirected split, with barely half the downstream energy in the corridor. Too strong and the optimizer suppresses the field rather than steering it: delivery collapses by more than an order of magnitude while confinement also falls, since a suppressed field is not a guided one. The intermediate value dominates the whole sweep on every metric simultaneously.

2 Periodic metamaterials: co-designed flat bands and band gaps

The wave-control problems above steer a wave through a finite plate. The same orthotropic fiber design also shapes the intrinsic *dispersion* of a periodic fiber-composite metamaterial. Here we co-design *two* fields on a single periodic unit cell — the density z (topology) and the fiber orientation θ (toolpath) — to make a chosen band *flat* (a slow-wave, high-density-of-states mode) and *isolated* by gaps to its neighbours. The constitutive model is identical to §**??** (the rotated orthotropic $\mathbf{D}_f(\theta)$ with SIMP density), but the solver is a different one.

Table 1: Effect of the leakage penalty λ_{out} on the joint-routing design (all other parameters fixed). Exit gain and guard ratio are relative to the straight-fiber baseline; a guard ratio below one means the fatigue-critical annulus is quieter than baseline. "Confinement" is the fraction of downstream ($x > x_J$) interior energy lying inside the corridor, which at the baseline is 22%.

λ_{out}	exit gain	guard ratio	protection	confinement
0.25	426×	0.13	3218×	45%
0.80	**698**×	**0.17**	**4002**×	**97**%
2.0	339×	0.15	2303×	99.9%
5.0	29×	0.29	99×	86%

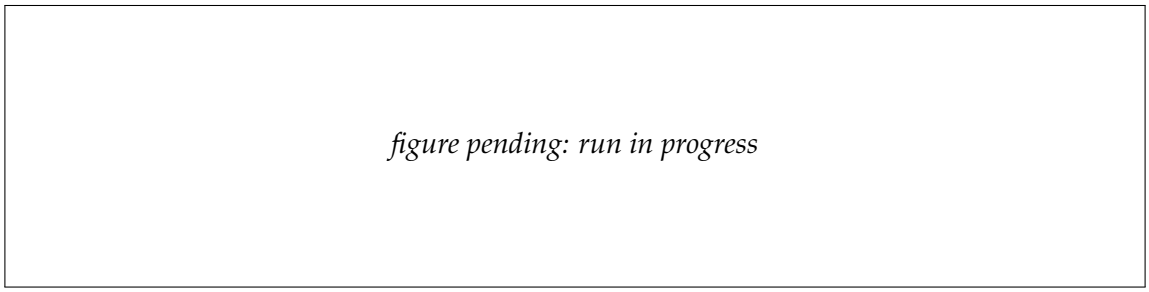


Figure 4: Routing elastic energy along a curved corridor around a through-hole joint. Top: the straight-fiber baseline, where energy concentrates on the joint, versus the orientation-guided design, which threads the wave along the corridor (cyan dashed) and into the exit port (light dashed box) while keeping the fatigue-critical guard annulus (red dotted) quiet. The joint is cut from the geometry, so its rim is exactly circular and traction-free. Bottom: the phase-field fiber toolpaths, which arc over the joint within the curvature limit $|\zeta| \le 1/R_J$, and the anisotropy director map. Only the fiber orientation is designed.

Remark 1 (Two independent solvers)**.** *The results of §**??**–§1 are computed in* `dolfinx/FEniCSx` *on an unstructured triangulation with* `PETSc` *linear algebra and UFL-generated adjoints (*`wavetopo/dolfinx_wave.py`*). The band-structure results of this section are* not*: they come from an independent, self-contained* `numpy/scipy` *Q4 quadrilateral code (*`wavetopo/cfrp.py`→`wavetopo/bloch.py`*) that imports no* `dolfinx`*,* `UFL`*,* `PETSc` *or* `SLEPc`*. Bloch periodicity is imposed by an explicitly constructed complex reduction matrix* $\mathbf{T}(\mathbf{k})$*, and (4) is solved as a* dense *Hermitian generalized eigenproblem (*`scipy.linalg.eigh` *on a band subset) — which is why the per-*$\mathbf{k}$ *cost grows steeply with mesh size (Table 2). The two codes therefore share the* physics *(the same* $\mathbf{Q}_f$*, the same* $\mathbf{D}_f(\theta) = \mathbf{T}\mathbf{Q}_f\mathbf{T}^{\mathsf{T}}$*, the same SIMP interpolation) and the same design machinery (CS-RBF map, density filter, MMA), but no assembly or linear-algebra code whatsoever. That independence is worth stating plainly rather than eliding: each implementation is a check on the other's constitutive algebra, and neither set of results inherits a bug from the other.*

2.1 Bloch–Floquet reduction and the two-field cell

On a rectangular cell of size $a \times b$ a Bloch wave obeys $\mathbf{u}(\mathbf{x}+\mathbf{R}) = e^{\mathrm{i}\mathbf{k}\cdot\mathbf{R}}\mathbf{u}(\mathbf{x})$ for lattice vectors $\mathbf{R}$. Assembling the full-cell stiffness and mass $\mathbf{K}(z,\theta)$, $\mathbf{M}(z)$ from the two-phase interpolation

$$\mathbf{K}_e = \big(\epsilon + (1-\epsilon)z_e^{\,p}\big)\mathbf{k}_f(\theta_e) + \big(1-z_e^{\,p}\big)\mathbf{k}_m, \qquad \mathbf{M}_e = \big(\rho_{\min} + z_e(1-\rho_{\min})\big)\rho_f\,\mathbf{m}_e \tag{3}$$

(stiff anisotropic fiber phase $\mathbf{k}_f(\theta)$ vs. soft isotropic matrix $\mathbf{k}_m$), Bloch periodicity is imposed by a complex reduction matrix $\mathbf{T}(\mathbf{k})$ that ties each boundary node to its master with the phase $e^{\mathrm{i}\mathbf{k}\cdot\mathbf{R}}$.

The reduced Hermitian generalized eigenproblem is

$$\mathbf{K}_r(\mathbf{k})\,\boldsymbol{\phi}_n = \omega_n^2(\mathbf{k})\,\mathbf{M}_r(\mathbf{k})\,\boldsymbol{\phi}_n, \qquad \mathbf{K}_r = \mathbf{T}^{\mathsf{H}}\mathbf{K}\,\mathbf{T}, \quad \mathbf{M}_r = \mathbf{T}^{\mathsf{H}}\mathbf{M}\,\mathbf{T}, \tag{4}$$

solved for the lowest bands along a Brillouin-zone path Γ–X–M–Γ. The anisotropy of $\mathbf{k}_f(\theta)$ is what makes the gaps *direction-dependent*, giving orientation genuine design authority over the dispersion.

2.2 Eigenvalue sensitivities and the flat-band objective

For a simple eigenvalue $\lambda_n = \omega_n^2$ with $\mathbf{M}_r$-orthonormal $\boldsymbol{\phi}_n$, first-order perturbation gives the design derivative

$$\frac{\partial \lambda_n}{\partial p} = \boldsymbol{\phi}_n^{\mathsf{H}}\Big(\frac{\partial \mathbf{K}_r}{\partial p} - \lambda_n \frac{\partial \mathbf{M}_r}{\partial p}\Big)\boldsymbol{\phi}_n, \qquad p \in \{z_e,\ \hat{\theta}_i\}, \tag{5}$$

which reduces to element energies of $\partial \mathbf{k}_f / \partial\theta$ (**??**) and the SIMP derivatives (`bloch.py : eigen_sensitivity`, FD-verified). To make band b flat and isolated we aggregate over the $\mathbf{k}$ set with Kreisselmeier–Steinhauser (KS) soft extrema and minimize

$$J = w_{\mathrm{f}} \underbrace{\big(\widetilde{\max}_k\,\omega_b - \widetilde{\min}_k\,\omega_b\big)}_{\text{bandwidth of } b} - w_{\uparrow} \underbrace{\big(\widetilde{\min}_k\,\omega_{b+1} - \widetilde{\max}_k\,\omega_b\big)}_{\text{gap above}} - w_{\downarrow} \underbrace{\big(\widetilde{\min}_k\,\omega_b - \widetilde{\max}_k\,\omega_{b-1}\big)}_{\text{gap below}} + c_v(\bar{z} - v^{\star})^2, \tag{6}$$

over z (density-filtered) and $\hat{\theta}$ (CS-RBF), by the same bound-constrained MMA (`wavetopo/flatband_opt.py`). Minimizing the bandwidth flattens b while the two gap terms push its neighbours away; all three share the $\widetilde{\max}_k \omega_b$ term consistently.

2.3 Result: an isolated flat band, and an as-manufactured check

So the tiled medium is physically connected — with mass on all four cell walls, not a set of disjoint struts — we fix a thin passive solid *frame* of fiber composite on the cell boundary (a locally-resonant frame enclosing a designed resonator, the classic flat-band morphology; `flatband_opt.py`, the `passive` mask). Co-optimizing the interior density and the fiber orientation then drives band 3 from a bandwidth of 2.78 to 0.59 (a 4.7× flattening) and opens isolating gaps on *both* sides (+0.42 above, +0.66 below), so the flat band sits alone in a complete gap (Figure 5). The optimized cell tiles seamlessly (periodic density and orientation) and is connected across every boundary, and both the topology and the fiber toolpaths are non-trivial — the flat band is a genuine *co-design* of where the material sits *and* which way its fibers run, neither alone sufficing.

To check manufacturability we re-simulate the *as-built* cell: the design density is projected to a crisp two-phase topology (the "cropped" material) and delineated into finite-pitch fiber tows (§**??**, masked to the solid), and the band structure is recomputed from that geometry and orientation. The flat band survives the crop — its bandwidth is 0.65 (still far below the baseline 2.78) and *both* isolating gaps remain (+0.41 above, +0.67 below; Figure 5, top right) — confirming the slow-wave mode and its complete gap are properties of the manufacturable part, not of the intermediate SIMP densities.

A caveat on the isolating gaps. The flat band itself is a *bandwidth* — a max minus a min of one band over the $\mathbf{k}$-sample — and is insensitive to sampling density, so the flattening is robust. Its *isolating gaps*, by contrast, are computed by the same discrete-$\mathbf{k}$ aggregation that, in the

multi-functional band-gap study (§3), reported complete gaps that closed under dense resampling. The gap widths quoted here should therefore be read as sampled values pending a dense-**k** re-evaluation; the flattening claim, which does not depend on the gap, stands regardless.

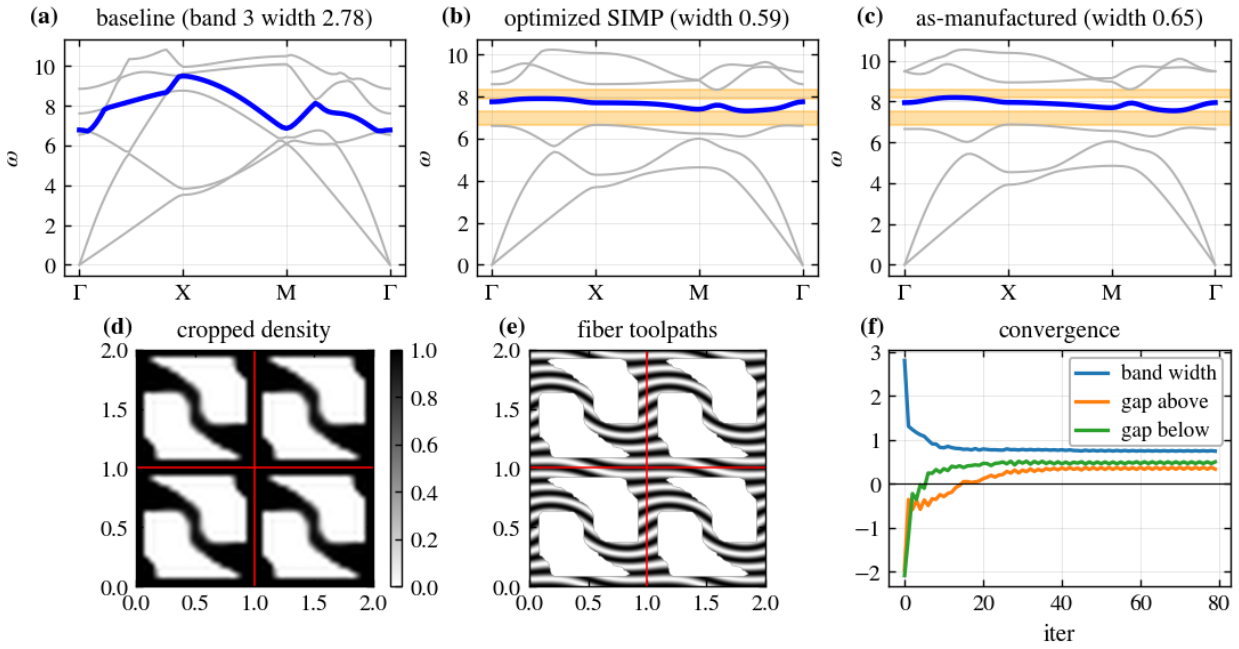


Figure 5: Flat-band + band-gap metamaterial by co-optimized topology and fiber toolpath, with an as-manufactured validation. Top: band structure (Γ–X–M–Γ) of the baseline, the optimized (SIMP) cell, and the *as-manufactured* cell (cropped density + delineated tows); the target band 3 (blue) is flattened from 2.78 to 0.59 and isolated by a gap (orange), and stays flat (0.65) with its upper gap (+0.41) under the manufactured geometry. Bottom: the crisp (projected) density; the as-manufactured fiber tows (phase-field wave projection, §**??**, masked to the solid and built from an exactly cell-periodic phase — the periodic part by FFT plus an integer number of tow cycles per period — so the tows join continuously across all four cell boundaries), each on a 2×2 tiling; and the convergence of the bandwidth and the two gaps. Design fields: density z *and* orientation θ.

2.4 Resolution and toolpath-pitch sweep: what extra design freedom buys

The design freedom in this problem is set by the orientation-support count $M = k_s^2$, the filter radius R, and the tow pitch d of (**??**). The analysis mesh n is *not* a design variable — it is what the physics is resolved on — so we hold it fixed at $n = 32$, along with the iteration budget, and vary only the parametrization. This matters: an earlier version of this experiment refined n together with (k_s, R, d), which forced N_k and the iteration count *down* as the design freedom went up (the n^6 cost of (**??**)), confounding the two effects beyond recovery. Table 2 reports the controlled comparison.

Design freedom helps — but only if the k-sampling keeps up. Rows C1–C3 share one **k**-set ($N_k = 22$), so they isolate the parametrization. Going from $M = 16$ to $M = 36$ improves the band as expected ($w_1 = 0.66 \to 0.56$). Going on to $M = 64$ (row C3) does not: the true width *degrades to* 0.98, worse than the coarsest design, and both isolating gaps nearly vanish. Row C3d

Table 2: **Controlled** design-freedom sweep for the flat-band cell (band $b = 3$). The analysis mesh ($n = 32$) and the iteration budget (70) are identical for every row, and C1–C3 additionally share one aggregation **k**-set ($N_k = 22$), so differences between them are attributable to the design parametrization alone: the orientation freedom $M = k_s^2$, the filter radius R, and the tow pitch d. **C3d repeats C3 with a denser k-set** ($N_k = 40$) and is the control for **k**-sampling. $w_0 \to w_1$ is the target-band width before and after optimization, measured on the common fine Γ–X–M–Γ path; the last three columns are the *as-manufactured* re-simulation (crisp projected density + delineated periodic tows at pitch d). Two entries carry the argument of §2.4: C3 is over-parametrized for its **k**-set and its true width (0.98) is *worse* than the coarsest design, while C3d — the same parametrization, sampled adequately — is the best row on every metric. C1's upper gap goes *negative* once manufactured, its coarse pitch $d = 1/6$ being unable to render the design faithfully.

	design freedom				sampling	optimized (SIMP, fine path)				as-manufactured		
run	k_s	M	R	d	N_k	w_0	w_1	$g_\uparrow$	$g_\downarrow$	w^{m}	$g_\uparrow^{\mathrm{m}}$	$g_\downarrow^{\mathrm{m}}$
C1	4	16	0.160	1/6	22	2.87	0.66	+0.16	+0.58	0.84	-0.14	+0.36
C2	6	36	0.120	1/9	22	2.52	0.56	+0.22	+0.45	0.65	+0.21	+0.52
C3	8	64	0.090	1/12	22	2.06	0.98	+0.13	+0.12	1.02	+0.08	+0.11
C3d	8	64	0.090	1/12	40	2.06	0.55	+0.53	+0.67	0.55	+0.59	+0.57

is the control — the *same* $M = 64$ parametrization, same mesh, same iterations, with the **k**-set enlarged to $N_k = 40$ — and it is the best row in the table on every metric ($w_1 = 0.55$, $g_\uparrow = +0.53$, $g_\downarrow = +0.67$). The finest parametrization was never the problem; sampling it with 22 wavevectors was.

The mechanism is *overfitting of the aggregation set*. The KS extrema in (6) constrain the band only at the N_k sampled wavevectors. With 64 design variables against 22 samples the design is over-parametrized, and the optimizer flattens precisely at the samples while the band excurses between them — the exact failure the discretization hierarchy (**??**) is meant to prevent, appearing in the **k**-domain rather than the spatial one.

Remark 2 (A free overfitting diagnostic)**.** *The KS bound* (**??**) *makes this detectable at no extra cost. Because* $\widetilde{\max}_\rho$ *can only* over*-estimate a maximum (and* $\widetilde{\min}_\rho$ *under-estimate a minimum), the aggregated bandwidth is an upper bound on the true width* over the sampled set*. If the width measured on a fine path nevertheless comes out* larger *than the aggregate, the band must be excursing between samples: the design has overfitted the* **k***-set. In Table 2 the aggregate widths at the final iterate are* 0.79, 0.70, 0.67 *and* 0.68 *for C1, C2, C3, C3d, against fine-path widths of* 0.66, 0.56, **0.98** *and* 0.55*. Only C3 has fine > aggregate, and C3 is exactly the row that fails. The test needs no extra optimization — only one fine-path evaluation of the converged design.*

Finer tows buy manufacturing fidelity. The as-manufactured columns separate the designs again, and along the axis one would expect: the tow pitch. C1, delineated at the coarse pitch $d = 1/6$, does not survive its own toolpath — its upper gap goes from $+0.16$ to $-\mathbf{0.14}$, i.e. the gap the design was optimized for *closes* once the fibers are rendered at finite spacing. C2 ($d = 1/9$) survives with both gaps intact, and C3d ($d = 1/12$) is essentially unchanged by manufacturing ($w_1 = 0.55 \to 0.55$, gaps $+0.53/+0.67 \to +0.59/+0.57$). Finer toolpath spacing is therefore not a cosmetic refinement: at $d = 1/6$ the manufactured part does not have the property the design was optimized for.

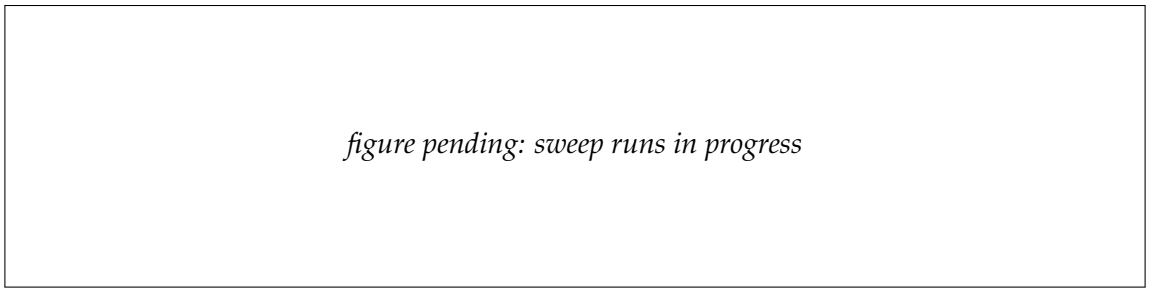


Figure 6: Effect of jointly refining the mesh n, the orientation-support grid k_s, and the tow pitch d. Columns: the three levels of Table 2. Top: band structure on the common fine path, with the target band 3 (blue) and the isolating gaps (orange). Middle: the crisp projected density on a 2×2 tiling — the finer levels resolve the resonator–frame ligaments that the coarse cell can only smear. Bottom: the as-manufactured tows at the corresponding pitch d, exactly cell-periodic and masked to the solid.

3 Multi-functional design: focusing and stiffness in one fiber layout

A structural panel often has to do two jobs at once: carry a mechanical load, and manage an elastic wave. This section poses both on a single finite part and asks whether one fiber layout can serve both — a genuine multi-functional co-design, on a finite domain, with *no band gap* anywhere.

Why not the periodic band-gap route. The natural first attempt is a periodic cell co-designed for homogenized stiffness and a complete phononic band gap. We pursued that and report its failure plainly, because it is instructive. The complete gap was aggregated over a discrete $\mathbf{k}$-sample during optimization; the optimizer drove the gap open *at the sampled points* while the bands crossed *between* them, so the "gaps" were artifacts of sparse $\mathbf{k}$-sampling. Re-evaluating the optimized cells on a dense 11×11 Brillouin-zone grid, the reported gaps of $+0.21$, $+0.39$ and $+0.55$ became -0.20, -0.09 and $+0.08$ — essentially closed — and straightening the fibers *widened* the true gap in every case, so the orientation was gaming the coarse metric rather than opening a gap. A band-gap claim is only as good as its $\mathbf{k}$-convergence, and this one did not survive it. We therefore move the multi-functional demonstration to a finite domain, where the objectives are integrals over the part rather than extrema over a Brillouin zone and no such sampling gap exists.

3.1 Two objectives on one solid plate

On the 4×3 solid orthotropic plate of §**??**, driven at $\omega = 40$ by the left-edge plane wave, the fiber orientation $\theta(\mathbf{x})$ is scored on

- **dynamic:** the focus gain $G = E_{\text{foc}}(\theta)/E_{\text{foc}}(0)$, the focus-region energy relative to straight fibers (the lens objective of §**??**, adjoint FD-verified to 6 digits);
- **static:** the cantilever compliance $c = \mathbf{F}_s^{\mathsf{T}}\mathbf{u}_s$ with $\mathbf{K}(\theta)\mathbf{u}_s = \mathbf{F}_s$ — the left edge clamped, a transverse load on the right edge. Compliance is self-adjoint, $\partial c/\partial\theta = -\mathbf{u}_s^{\mathsf{T}}(\partial\mathbf{K}/\partial\theta)\mathbf{u}_s$, FD-verified to 2.9×10^{-7}.

Both are functions of the same θ on the same plate. The tension is real: the converging-lens layout that bends the wave toward the target is not the load-aligned layout that resists bending.

3.2 Result: one layout, both functions

Three designs from an identical straight-fiber start make the point. The *focus-only* lens reaches a focus gain of 46× but drops the panel stiffness to 0.73× that of straight fibers — steering the wave costs 27% of the structural stiffness. Constraining the compliance to the straight-fiber level and maximizing focus (augmented Lagrangian on the single compliance constraint, as in §**??**) gives the *co-design*: a focus gain of 22× at 1.02× the straight-fiber stiffness — essentially all of the stiffness *and* a large fraction of the focusing, from one fiber field.

Sweeping the compliance limit traces the focus–stiffness front: focus gain rises from 22× at full stiffness, through 35× at a 20% compliance allowance, to the focus-only endpoint, saturating once the stiffness constraint is loose enough to stop binding. The designer reads an operating point off this curve rather than choosing between the two functions. Unlike the periodic band-gap result, every point here is an integral over the finite part and carries no **k**-sampling caveat.

4 Wave shielding on a solid CFRP plate with drilled holes

The multi-functional result of §3 designs the fiber orientation alone, on a solid homogeneous plate. A real composite panel adds one feature the orientation-only picture omits: *prescribed through-holes* — bolt holes, fastener passages, access cut-outs — whose rims are traction-free boundaries the wave scatters from. Here we turn that feature into an asset. Exactly as the elastic cloak (§**??**) wraps the fiber toolpath around a real hole to control its scattered field, we design the toolpath of a *holed* panel to *shield* sensitive regions: harmonic vibration enters at *both* bottom corners — two independent source load cases — and the fiber layout must divert it away from *both* of the opposite top corners, across a *band* of frequencies, while the holes cut weight and the fiber alignment keeps the panel stiff.

Insisting on *real* holes here — cut from the geometry, with no material inside — rather than soft-density voids is decisive for a wave objective. A wave localizes in the slow, compliant phase, so a soft-density void *attracts* the field: any weight saved by softening material reappears as wave energy sitting in what looks like a void, and a mass objective and a wave objective end up fighting over the same cells. A traction-free hole does the opposite — the wave reflects off its rim and cannot enter — so the shielded field lives entirely in solid material, there is provably zero wave energy in any void, and mass reduction and wave control stop competing. This is the cloak's mechanism (orientation control around real holes), aimed at a protected region instead of at invisibility.

4.1 A conforming vector element: the missing discretization

Every in-plane (vector) result in this paper so far lives on a structured quad grid; every result on a hole-conforming `gmsh` triangulation is scalar antiplane (§**??**). This problem needs *both at once* — vector elasticity on a mesh that conforms to holes — so we add the element that was missing: a constant-strain-triangle (CST) plane-stress element carrying the same rotated orthotropic fiber stiffness $\mathbf{D}_f(\theta)$ as the quad elements. It reuses, unchanged, the exact five-term Fourier decomposition of Appendix **??**: with $\mathbf{B}$ the 3×6 CST strain-displacement matrix and A the triangle area,

$$\mathbf{k}_f(\theta) = \sum_{s=0}^{4} c_s(\theta) \underbrace{A\,\mathbf{B}^{\mathsf{T}}\mathbf{C}_s\,\mathbf{B}}_{\text{5 fixed } 6\times 6 \text{ matrices}}, \qquad \mathbf{c}(\theta) = \big[1,\ \cos 2\theta,\ \sin 2\theta,\ \cos 4\theta,\ \sin 4\theta\big], \tag{7}$$

so element rotation is five precomputed matrices weighted by trigonometric scalars, and $\partial \mathbf{k}_f / \partial \theta$ costs only the derivative of $\mathbf{c}(\theta)$. The mesh is built by cutting the disks from the rectangle in the

OCC kernel, giving *exactly* circular, traction-free rims (a natural boundary condition), graded finer toward each rim — never a soft-density staircase, for the scattering reasons argued in §6.2. The Fourier fit is exact to 8.5×10^{-16}, and both design adjoints (the dynamic protected-energy and the static compliance, each w.r.t. θ) pass the finite-difference check to $\lesssim 10^{-6}$.

4.2 Result: a shielded region, entirely in solid material, at lower mass

On a 7×4 panel carrying five drilled through-holes (radii 0.48–0.60), meshed conformally, harmonic vibration enters at *both* bottom corners — two independent source load cases — and we ask that it stay out of *both* top corners across a band of frequencies $\omega \in \{9, 11, 13\}$ (Figure 7, Table 3). With straight fibers, 7–12% of the interior energy reaches the protected corners at each frequency. Minimizing the band-averaged protected energy — summed over both sources, with one LU factorization per frequency serving every source load case — over the fiber orientation, with a printable-curvature penalty and a compliance term and only the orientation as a variable, diverts the wave: the protected fraction falls to 2–3%, a 3–5$\times$ reduction *at every frequency*, with both corners dark across the whole band while the field stays brightest at the two sources. Every bright region is solid, and every hole shows exactly zero energy inside, because the wave cannot enter a real void. The same layout is 24% stiffer than straight fiber and, through the five holes, 15% lighter than the full plate, at a maximum fiber curvature $|\zeta|_{\max} = 1.1$ — inside the $1/R$ bound of the smallest hole, so the tows are printable. The orientation gradient is finite-difference verified to $\lesssim 10^{-6}$ and, by the isotropy-annihilation identity (Remark **??**), driven purely by the fiber anisotropy: setting $\mathbf{Q}_f$ isotropic sends every $\partial E/\partial\theta$ to $O(10^{-16})$.

Table 3: Broadband, two-source, multi-target fiber-orientation wave shield on the holed CFRP panel: the share of interior energy reaching the union of the two protected top corners (summed over the two bottom-corner source load cases), straight fibers versus the optimized toolpath, at each frequency. The single optimized layout is 0.76$\times$ the straight-fiber cantilever compliance (24% stiffer) at a maximum fiber curvature $|\zeta|_{\max} = 1.1$ (printable). Both designs carry the same five drilled holes, so both are 0.85$\times$ the solid-plate mass, and by construction neither has any wave energy in a void.

frequency ω	straight fraction	optimized fraction	quieter
9	0.070	**0.022**	**3**$\times$
11	0.119	**0.026**	**5**$\times$
13	0.117	**0.029**	**4**$\times$

This is the cloak's anisotropy lever (§**??**) aimed at a protected region rather than at invisibility, and it settles the tension that a soft-void topology cannot: because the removed material is real holes, not compliant voids, the wave has nowhere soft to collect, so the panel is lighter *and* the shielded field never leaves the solid. Nothing in the objective is periodic and every quantity is an integral over a finite, buildable part.

5 A stiffer, stronger bracket with a higher fundamental: topology and toolpath for a modal objective

The shield of §4 designs a toolpath on a fixed hole pattern. Our last finite-part example designs *both* the topology and the toolpath, and targets a *modal* objective, using the eigenvalue sensitivity

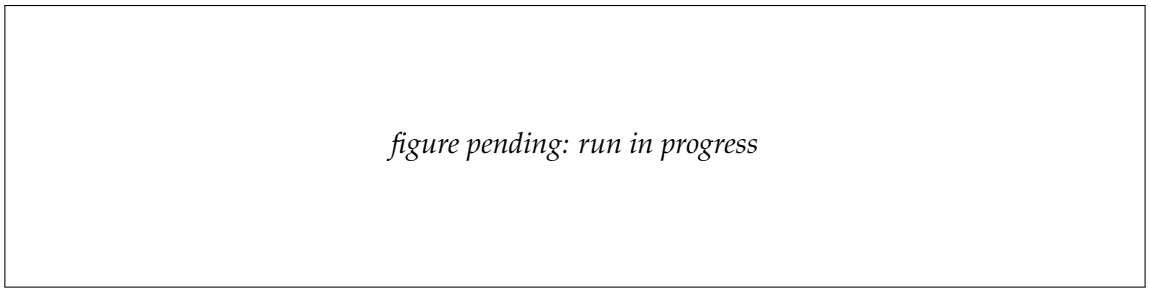


Figure 7: Broadband, two-source, multi-target fiber-orientation wave shield on a solid CFRP plate with five drilled through-holes. Vibration enters at *both* bottom corners (yellow) and must be kept out of *both* top-corner regions (green dashed) across $\omega \in \{9, 11, 13\}$. (a) The optimized fiber toolpath, drawn as tows on the solid with the holes cut out; $0.85\times$ the solid-plate mass. (b) The fiber curvature $|\zeta|$ (max 1.1, everywhere inside the $1/R$ print limit). (c) The protected-corner energy fraction at each frequency, straight fibers versus optimized: the shield reduces it 3–5$\times$ across the band. (d–f) The combined field over the two sources at $\omega = 9, 11, 13$: both corners stay dark at every frequency, the field is brightest at the sources, and every hole is a real void with exactly zero wave energy inside — the shielded field is entirely in solid material.

that drives the metamaterial of §2 — here on a clamped finite part rather than a Bloch cell. The part is a mounting bracket: bolted at two holes on the clamped root and carrying a service load on the rim of a third through-hole at the tip, it should have the highest possible first natural frequency ω_1 — to keep the fundamental resonance clear of the operating excitation — at a *fixed mass*. Maximizing a single eigenvalue in isolation is a trap: because a Rayleigh quotient rewards any mass placed where it raises stiffness faster than inertia, the optimizer readily grows *neutral outriggers* — stubby cantilevers near the root that nudge ω_1 up by a hair but carry no service load, so the "bracket" sprouts dead appendages (removing them changes ω_1 by under half a percent but they persist in every volume-constrained run, regrowing even when their region is forbidden). We therefore co-design against *three* objectives at once, all at fixed mass: maximize ω_1; minimize the static compliance, so every member is load-bearing rather than a frequency outrigger; and minimize the peak stress, so the load path is thick and uniformly stressed rather than a set of thin overloaded ligaments. It is a single solid anisotropic material — no fiber/matrix blend, only fiber and true void (a 1% Ersatz keeps the operators non-singular) — so "mass" here counts only real material, the compliance and the p-norm von Mises peak stress (Appendix .5) are both taken under the tip load, and each carries its own adjoint.

The counterintuitive lever. It is tempting to read "higher ω_1" as simply "stiffer." But ω_1^2 is a Rayleigh quotient $\boldsymbol{\phi}^\mathsf{T}\mathbf{K}\boldsymbol{\phi}/\boldsymbol{\phi}^\mathsf{T}\mathbf{M}\boldsymbol{\phi}$, and its density sensitivity (23) carries two terms of opposite sign: a positive stiffness term that wants material where the mode strains, and a *negative* mass term, $-\lambda_1(\rho_f - \rho_m)\,\boldsymbol{\phi}_e^\mathsf{T}\mathbf{m}_0\boldsymbol{\phi}_e$, that wants material *removed* where the mode has kinetic energy — at its antinodes. At a fixed mass budget the optimizer plays these against each other: it draws material toward the strained root and *scoops it out of the antinode at the tip*, so the fundamental rises far more than the change in total stiffness alone would suggest. The compliance and peak-stress objectives keep that redistribution physical and honest: compliance forces the redistributed material onto genuine load paths rather than the neutral root outriggers a bare eigenvalue objective would grow, and the peak-stress term spreads those paths into thick, uniformly stressed members rather than thin overloaded ligaments.

The design we build. We solve the bracket on a fine 3.5×2 mesh ($\sim$68k elements), enforcing mirror symmetry about the mid-height axis, clamp-connectivity (only material grounded to the support survives, so there are no floating islands), and thin passive-solid collars backing every hole; every ratio is against an *equal-mass* uniform-density plate. The manufacturable result is the clean, fully load-bearing A-frame of Figure 8: a top chord in tension and a bottom chord in compression sweeping from the clamped root to the loaded tip ring, the fiber tows wrapping every hole and following each chord. The tension/compression map (Figure 8b) shows *every member carries load* — there are no dead appendages. It is $7.3\times$ stiffer than the equal-mass plate, holds the peak stress to $0.15\times$, and raises the fundamental to $2.83\times$; that rise is *proportionate* (ω_1^2 tracks the stiffness gain) — the honest reading of "higher ω_1 means stiffer." The design is binary ($M_{\text{nd}} = 0.02$) and printable: an adaptive curvature penalty drives $|\zeta|_{\max}$ right up to the AFP limit $1/R$ — the tows use the *full* manufacturability budget to follow the load — and no further.

The frequency lever, and why we do not build it. The modal objective can push the fundamental much higher, but not for free. Sweeping the modal weight w_ω from 0 to 1 traces the stiffness–frequency evolution of Figure 9: the pure-stiffness A-frame sits *on* the proportionate line $\omega_1^2 \propto$ stiffness, and the first touch of frequency weight lifts the design far above it into the *disproportionate* region — ω_1 climbs toward 3.8–$4\times$, so ω_1^2 rises $\sim$14–$16\times$ against only a $\sim 5\times$ stiffness gain, the extra factor being mass scooped off the mode's tip antinode via the $-\lambda_1\, \partial\mathbf{M}/\partial z$ term (against a *fully solid* plate the frequency-weighted design more than doubles ω_1 at $0.42\times$ its mass). But that extra frequency is bought with lightly-loaded *modal* material: the frequency-weighted designs grow an inner arrow and stubby outriggers that a tension/compression check shows carry almost no service load — they reshape the mode, not the load path. Trimming them costs under 2% of ω_1 and returns the A-frame. So the lever is real and physically interesting, but the load-bearing, manufacturable part is the clean A-frame — the sweep is how we know exactly what the lever would cost to build.

Mode separation by anisotropy. The same eigenvalue machinery does something the load path cannot: play ω_1 against ω_2. Maximizing ω_1 *while minimizing* ω_2 — with a second-mode eigenvalue adjoint (Appendix .4) alongside the first — drives the two lowest natural frequencies *together*: the gap ω_2/ω_1 collapses from $\sim$4 toward $\sim$1.2 as the mode-2 weight w_2 is raised (Figure 10), tuning the bracket toward a near-degenerate fundamental pair. It is a purely anisotropic effect — the fiber field stiffens the first mode's strain field while softening the second's, which an isotropic redesign at fixed mass cannot do as sharply — and a stringent test of the sensitivities near an eigenvalue crossing.

Designed anisotropy is doing the work. The fiber orientation is a full design field, not a fixed layup. On the frozen A-frame topology the optimized tows are 44% stiffer than the best straight, single-angle laminate: the anisotropy is designed, not incidental. The tows run along the chords and wrap every hole (Figure 8a), putting the $E_{f1} = 131$ GPa axis on the load to within $\sim 6^\circ$ of the principal stress on every loaded member. The rotated stiffness $D_f(\theta) = \mathbf{T}\mathbf{Q}_f\mathbf{T}^{\mathsf{T}}$ reproduces the analytical off-axis modulus $E_x(\theta)$ — from 131 GPa along the fiber to 9 GPa across — to machine precision, and the finer the fiber field is resolved the harder the tows follow the load: a coarse support grid flattens them artificially, leaving $\sim$30% of the anisotropy on the table. Tellingly, the *modal* objective reaches for anisotropy hardest — left unpenalized its tows would curl to $|\zeta| \approx 14$ versus ≈ 2 for the compliance design, so the printability penalty binds on the frequency design first. All sensitivities — the two eigenvalue sensitivities for ω_1 *and* ω_2 (density and orientation,

§.4), the compliance adjoint, and the p-norm stress adjoint (§.5) — are finite-difference verified to $\lesssim 10^{-6}$, and every orientation gradient isotropy-annihilates.

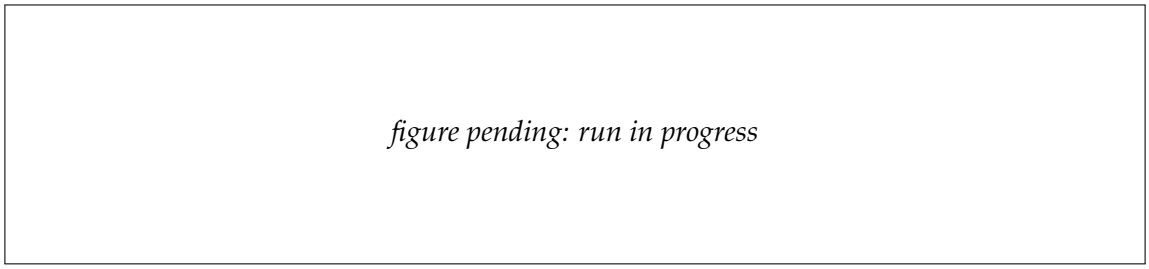


Figure 8: The fully-resolved CFRP mounting bracket (stiffness/compliance/stress co-design, ∼68k elements): a clean, fully load-bearing A-frame at equal mass. Bolted at two root holes (clamp edge, red), loaded on the tip-hole rim (red); thin passive-solid collars back every hole. (a) Topology with the continuous fiber tows, which *wrap every hole* and follow each chord. (b) The load path (red tension, blue compression): a top tension chord and a bottom compression chord — *every* member carries load, with no dead appendages. (c) Fiber curvature $|\zeta|$, driven right to the print limit $1/R$ so the tows exploit the full manufacturability budget. (d) The fundamental mode, $\omega_1 = 2.83\times$ the equal-mass plate. The design is $7.3\times$ stiffer and holds the peak stress to $0.15\times$; its optimized fibers are 44% stiffer than any straight layup.

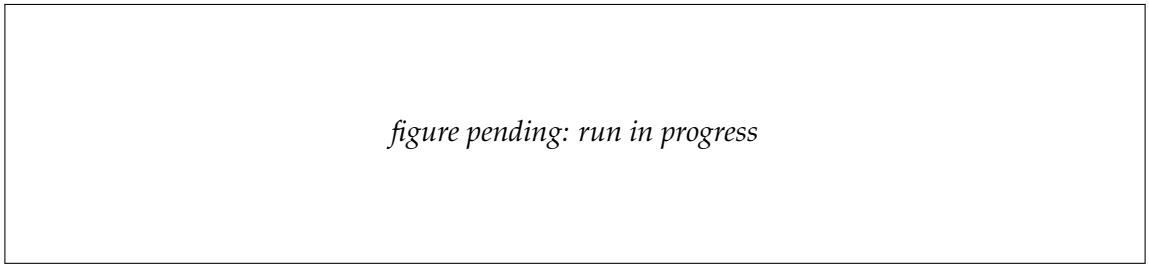


Figure 9: The frequency lever, and what it costs. Sweeping the modal weight w_ω from 0 (the manufacturable A-frame of Figure 8) to 1 (frequency-weighted) traces the stiffness–frequency evolution at equal mass. *Top:* the part morphs from the clean A-frame into a thinner truss with an inner arrow and outriggers as w_ω rises. *Bottom:* the pure-stiffness design lands on the proportionate line $\omega_1 = \sqrt{\text{stiffness}}$ (dashed); any frequency weight lifts it into the *disproportionate* region above — ω_1 reaches $3.8\text{–}4\times$ ($\omega_1^2 \sim 14\text{–}16\times$ the stiffness gain). The catch is that the extra material is a lightly-loaded modal stiffener, not a load path, so the design we build is the $w_\omega{=}0$ end.

6 Topological valley guiding, and what the mesh must resolve

The orientation lever of §**??** steers waves through a homogeneous plate; the periodic cell of §2 shapes a dispersion relation. A third regime combines them: a *topological* valley-Hall waveguide, in which the orientation of a C_3 scatterer acts as a tunable Dirac mass and a domain wall between mirror-partner orientations guides a kink mode. This section reports that device, and — because the result turns on it — the discretization requirement that makes it meaningful.

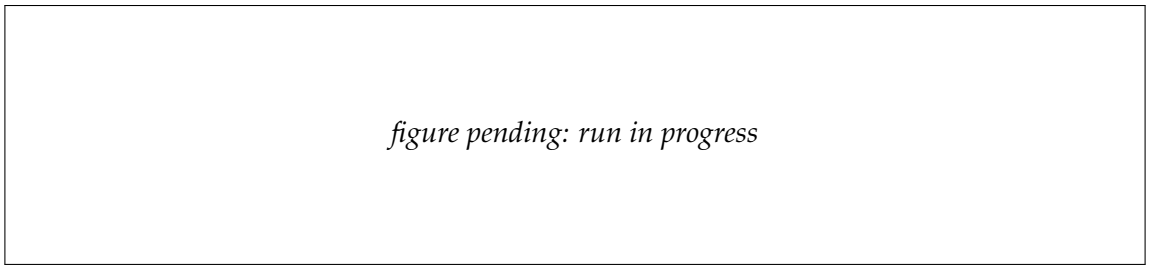


Figure 10: Mode separation by designed anisotropy: maximizing ω_1 *while minimizing* ω_2 drives the first two natural frequencies together at fixed mass. *Top:* topology and fiber tows as the mode-2 weight w_2 is raised. *Bottom left:* the gap ω_2/ω_1 collapses from $\sim$4 toward $\sim$1.2 — a near-degenerate fundamental pair. *Bottom middle/right:* the first two mode shapes of the most mode-merged design; the fiber field has tuned them to nearly coincide. This is a purely anisotropic effect and a stringent check of the second-mode sensitivity.

6.1 The crystal and the protected window

We use the scalar (antiplane-shear) problem on a triangular lattice ($\mathbf{a}_1 = a(1,0)$, $\mathbf{a}_2 = a(\frac{1}{2}, \frac{\sqrt{3}}{2})$) with one stiff equilateral-triangle rod per cell ($R = 0.44a$, $\mu_f/\mu_m = 100$). An equilateral triangle is C_{3v}; aligning its mirrors with the lattice's gives a symmetry-protected Dirac cone at K, and rotating the rod by α breaks the mirror while preserving C_3, opening a valley gap with Dirac mass $m \propto \cos(3\alpha)$. Mirror-partner domains (α and $60^\circ - \alpha$, here $15^\circ|45^\circ$) carry opposite m, hence $|\Delta C_v| = 1$, and their interface hosts a kink mode.

Three independent calculations, all on *rod-conforming* meshes (§6.2), bound where that mode may be claimed as topological. The complete bulk gap from the periodic cell (`wavetopo/valley_cell.py`, converged in Brillouin-zone sampling: the band-2 maximum is stable to four decimals and the band-3 minimum moves by 0.019 between 169 and 1681 $\mathbf{k}$-points) is

$$\omega \in [7.014,\ 7.567], \qquad \text{width } 0.553. \tag{8}$$

The ribbon supercell with a $15^\circ|45^\circ$ wall carries a strongly wall-localized branch spanning $\omega \approx 6.5$ to 7.27. The *protected window* is the intersection — the wall branch lying inside the complete gap:

$$\omega \in [7.014,\ 7.27]. \tag{9}$$

Guiding is broader than protection. In the finite domain the wall guides well over roughly $[7.05, 7.85]$: the field hugs the wall and reaches a probe at the far end, with wall confinement $C = \int_{d<1} |u|^2 / \int_{\text{int}} |u|^2 = 0.95$ (against 0.51 below the gap). But confinement stays high to $\omega \approx 7.85$, well *above* the complete gap edge 7.567. Fields at $\omega = 7.7$ confirm genuine, tightly confined transport. It is nonetheless not topological: above 7.567 bulk states exist and the ribbon carries no wall branch, so what guides there is conventional index contrast at the wall, not a gap-traversing kink mode. We therefore restrict every protection claim to (9) and report the 7.6–7.85 guiding separately as ordinary guiding. Distinguishing the two requires the confinement measure: transmission to a probe registers energy that arrived by *any* route and reads high in both regimes, differing by only a factor $\sim$3.

6.2 Why the mesh must conform to the rods

The natural finite-domain discretization — mesh the rectangle, then assign each element to rod or matrix by testing its centroid — is the staircased construction rejected for through-holes in §1 and for the cloaked void in §**??**. Here it is not merely inaccurate but *disqualifying*, because the Dirac mass $m \propto \cos(3\alpha)$ *is* the C_3 symmetry breaking of the scatterer: a staircase that perturbs the rod's threefold symmetry by an amount comparable to the intended mass perturbs the gap itself.

The perturbation is not small. Painting rods onto a plain mesh at $h = 0.09$ leaves the *mean* rod area correct to 0.09% — staircase errors average out over 293 rods, so a bulk fiber-fraction check does not detect the problem — while giving each individual rod a random area error and displacing its effective centre by up to 2.9% of R. Cutting the rods into the geometry instead reproduces every rod exactly: per-rod area 0.251487 with standard deviation 2.2×10^{-16} and centroid offset 3×10^{-16}.

Table 4 shows the consequence. Sweeping wall transmission over frequency at three resolutions, the conforming meshes agree to a factor 1.9 and peak inside the gap (8) at every resolution. The painted meshes spread by a factor 590, and at the coarsest resolution their peak falls at $\omega = 8.0$, outside the gap altogether — no guided band at all. Painted meshes do converge toward the conforming answer, but slowly: at $h = 0.06$ they are still a factor ~ 4 low. The conforming construction is converged at $h = 0.12$, where the painted one is not converged at half that spacing.

Table 4: Mesh-convergence of the guided band. $\bar{T}$ is the mean transmission over the complete gap (8); ω_{peak} is where transmission is largest.

mesh	h	$\bar{T}$ (in gap)	ω_{peak}	in gap?
conforming	0.12	1.97×10^{-1}	7.267	yes
conforming	0.09	3.71×10^{-1}	7.200	yes
conforming	0.06	2.99×10^{-1}	7.133	yes
painted	0.12	1.34×10^{-4}	8.000	**no**
painted	0.09	1.38×10^{-2}	7.467	yes
painted	0.06	7.88×10^{-2}	7.533	yes

6.3 Robustness to wall defects

Within the protected window we damage the wall and re-measure. At $\omega = 7.2$ on a conforming mesh, transmission falls from 0.65 (pristine) to 21% of that value for a removed rod and 38% for a cluster of rods rotated by $\pm 12^\circ$, while all three remain more than three orders of magnitude above the out-of-band floor (pristine contrast $5400\times$).

Two qualifications keep this honest. First, this is degradation, not immunity: a vacancy costs a factor ~ 5. Valley-Hall protection suppresses *inter-valley* backscattering, and a vacancy is a sharp defect that supplies exactly the large momentum transfer the mechanism does not forbid; the correct claim is that the mode survives damage that would extinguish a resonant defect line, not that it is immune. Second, a defect test is only meaningful if the pristine transmission is well clear of the floor — otherwise all three configurations report noise and "survives damage" is vacuous. The driver therefore refuses to run unless the pristine in-band/out-of-band contrast exceeds a threshold. An earlier version of this study, on a painted mesh and at a frequency read off a painted sweep, returned 4.2×10^{-6}, 4.4×10^{-6} and 1.2×10^{-5} for the three configurations: all at the deep-gap floor, and no conclusion about protection follows from them.

.1 Exactly periodic phase on a unit cell

On the metamaterial cell the construction above must be modified: no node may be pinned, because ψ is not periodic — it must advance by a whole number of tow cycles per lattice vector, or the tows will not join across the boundary. Split

$$\Psi(\mathbf{x}) = \psi_p(\mathbf{x}) + n_x \frac{x}{a} + n_y \frac{y}{b}, \qquad \psi_p \text{ cell-periodic}, \quad (n_x, n_y) \in \mathbb{Z}^2. \tag{10}$$

The periodic part solves $\min_{\psi_p} \int_\Omega \|\nabla\psi_p - \mathbf{g}/d\|^2$, which is *diagonal* in the Fourier basis: with $\psi_p = \sum_{\mathbf{k}} \hat{\psi}_{\mathbf{k}} e^{\mathrm{i}\mathbf{k}\cdot\mathbf{x}}$ and $\nabla \mapsto \mathrm{i}\mathbf{k}$, the normal equations $\mathrm{i}\mathbf{k} \cdot \overline{(\mathrm{i}\mathbf{k}\hat{\psi}_{\mathbf{k}})} = \mathrm{i}\mathbf{k} \cdot \hat{\mathbf{g}}_{\mathbf{k}}/d$ give

$$\boxed{\hat{\psi}_{\mathbf{k}} = -\mathrm{i}\,\frac{\mathbf{k}\cdot\hat{\mathbf{g}}_{\mathbf{k}}}{d\,\|\mathbf{k}\|^2} \;\; (\mathbf{k} \neq \mathbf{0}), \qquad \hat{\psi}_{\mathbf{0}} = 0} \tag{11}$$

— the inverse divergence (Helmholtz projection) of $\mathbf{g}/d$, one FFT pair, no linear solve. The $\mathbf{k} = \mathbf{0}$ mode is precisely the part no periodic function can carry, so the *mean* gradient must be absorbed by the linear term; matching it in the least-squares sense and rounding to the nearest representable integer,

$$n_x = \mathrm{round}\Big(\frac{a\,\langle g_x\rangle}{d}\Big), \qquad n_y = \mathrm{round}\Big(\frac{b\,\langle g_y\rangle}{d}\Big). \tag{12}$$

The rounding is what buys exact periodicity: under $x \mapsto x + a$ the phase advances by $n_x \in \mathbb{Z}$, so $\chi = \frac{1}{2} + \frac{1}{2}\cos(2\pi\Psi)$ is *unchanged*, and the tows join continuously across every cell boundary.

The price is a slight, uniform stretch of the pitch. Since rounding perturbs the cycle count by at most $\frac{1}{2}$ out of $n_{\mathrm{tow}} \approx a/d$ cycles, the relative pitch error is bounded by

$$\frac{|\Delta d|}{d} \;\leq\; \frac{1}{2\,n_{\mathrm{tow}}} \;\approx\; \frac{d}{2a}. \tag{13}$$

This is a second, independent reason to prefer a fine tow pitch in §2.4: the coarsest level ($d = 1/6$) can suffer up to 8.3% pitch distortion in enforcing periodicity, the finest ($d = 1/12$) at most 4.2%. Exact periodicity and faithful pitch are competing goals, reconciled only by taking $d \ll a$.

.2 Properties of the aggregation and symmetry operators

KS bounds (??). Since every term of $\sum_k e^{\rho v_k}$ is positive and one of them is $e^{\rho \max_k v_k}$, $\sum_k e^{\rho v_k} \geq e^{\rho \max_k v_k}$, so $\rho^{-1} \ln \sum_k e^{\rho v_k} \geq \max_k v_k$. Conversely every term is at most $e^{\rho \max_k v_k}$, so $\sum_k e^{\rho v_k} \leq N_k e^{\rho \max_k v_k}$ and $\rho^{-1} \ln \sum_k e^{\rho v_k} \leq \max_k v_k + \rho^{-1} \ln N_k$. Hence KS *over*-estimates a maximum and *under*-estimates a minimum: the aggregated bandwidth in (6) is an upper bound on the true sampled bandwidth, and the aggregated gaps are lower bounds. Both errors are conservative — the optimizer is never rewarded for an artefact of the smoothing — and both vanish as $\rho^{-1} \ln N_k$. □

.3 Eigenvalue sensitivity for the Bloch metamaterial

The band-structure objectives of §2 are functions of the eigenfrequencies $\omega_n(\mathbf{k})$, so the optimizer needs $\partial\omega_n/\partial p$ for each design variable p. Computing this by finite differences would cost one extra eigensolve per variable per $\mathbf{k}$-point — thousands of eigensolves per iteration — so we instead use the analytic first-order perturbation result below, which delivers every $\partial\lambda_n/\partial p$ from quantities *already* computed by the single eigensolve at that $\mathbf{k}$. This is the discrete counterpart of the Hellmann–Feynman theorem of quantum mechanics (the energy derivative of a Hamiltonian equals the expectation of the Hamiltonian's derivative in the unperturbed state); the derivation makes precise why no eigenvector derivative survives.

Setup. At a fixed wavevector $\mathbf{k}$ the reduced problem (4) is the generalized Hermitian eigenproblem

$$\mathbf{K}_r\,\boldsymbol{\phi}_n = \lambda_n\,\mathbf{M}_r\,\boldsymbol{\phi}_n, \qquad \lambda_n = \omega_n^2, \qquad \mathbf{K}_r = \mathbf{K}_r^{\mathsf{H}},\ \ \mathbf{M}_r = \mathbf{M}_r^{\mathsf{H}} \succ 0, \tag{14}$$

with $\mathbf{K}_r, \mathbf{M}_r$ depending on the design $p \in \{z_e, \hat{\theta}_i\}$ through the assembled cell matrices. We assume λ_n is a *simple* (non-degenerate) eigenvalue — the caveat when it is not is discussed at the end — and fix the scale of the eigenvector by the $\mathbf{M}_r$-normalization

$$\boldsymbol{\phi}_n^{\mathsf{H}}\mathbf{M}_r\boldsymbol{\phi}_n = 1. \tag{15}$$

Write $(\cdot)' \equiv \partial(\cdot)/\partial p$. The matrices, the eigenvalue, and the eigenvector all depend on p; the goal is λ_n' alone.

Differentiate the eigen-relation. Applying $\partial/\partial p$ to (14) by the product rule gives

$$\mathbf{K}_r'\boldsymbol{\phi}_n + \mathbf{K}_r\boldsymbol{\phi}_n' = \lambda_n'\,\mathbf{M}_r\boldsymbol{\phi}_n + \lambda_n\mathbf{M}_r'\boldsymbol{\phi}_n + \lambda_n\mathbf{M}_r\boldsymbol{\phi}_n'. \tag{16}$$

This still contains the *unknown* eigenvector derivative $\boldsymbol{\phi}_n'$, which is expensive (it requires solving a singular system, or summing over all other modes). The trick is to project it out. Left-multiply (16) by $\boldsymbol{\phi}_n^{\mathsf{H}}$:

$$\underbrace{\boldsymbol{\phi}_n^{\mathsf{H}}\mathbf{K}_r'\boldsymbol{\phi}_n}_{\text{(A)}} + \underbrace{\boldsymbol{\phi}_n^{\mathsf{H}}\mathbf{K}_r\boldsymbol{\phi}_n'}_{\text{(B)}} = \lambda_n' \underbrace{\boldsymbol{\phi}_n^{\mathsf{H}}\mathbf{M}_r\boldsymbol{\phi}_n}_{=1} + \lambda_n \underbrace{\boldsymbol{\phi}_n^{\mathsf{H}}\mathbf{M}_r'\boldsymbol{\phi}_n}_{\text{(C)}} + \lambda_n \underbrace{\boldsymbol{\phi}_n^{\mathsf{H}}\mathbf{M}_r\boldsymbol{\phi}_n'}_{\text{(D)}}. \tag{17}$$

Why the eigenvector-derivative terms cancel. Consider the two terms carrying $\boldsymbol{\phi}_n'$, namely (B) and (D). For (B), use that $\mathbf{K}_r$ is Hermitian and $\boldsymbol{\phi}_n$ is its eigenvector: taking the conjugate transpose of $\mathbf{K}_r\boldsymbol{\phi}_n = \lambda_n\mathbf{M}_r\boldsymbol{\phi}_n$ and using $\mathbf{K}_r^{\mathsf{H}} = \mathbf{K}_r$, $\mathbf{M}_r^{\mathsf{H}} = \mathbf{M}_r$, $\lambda_n \in \mathbb{R}$ (real, since the pencil is Hermitian-definite) gives the *left* eigen-relation

$$\boldsymbol{\phi}_n^{\mathsf{H}}\mathbf{K}_r = \lambda_n\,\boldsymbol{\phi}_n^{\mathsf{H}}\mathbf{M}_r. \tag{18}$$

Substituting (18) into (B),

$$\text{(B)} = \boldsymbol{\phi}_n^{\mathsf{H}}\mathbf{K}_r\boldsymbol{\phi}_n' = \lambda_n\,\boldsymbol{\phi}_n^{\mathsf{H}}\mathbf{M}_r\boldsymbol{\phi}_n' = \lambda_n\,\text{(D)},$$

so on the right-hand side the term $\lambda_n \cdot$ (D) is *identical* to (B) on the left, and the two cancel when (17) is rearranged. The unknown $\boldsymbol{\phi}_n'$ has dropped out entirely — this is the crux, and it is exactly the discrete Hellmann–Feynman cancellation: the first-order change of a normalized eigenvector is $\mathbf{M}_r$-orthogonal to the mode itself, so it cannot change the Rayleigh quotient to first order. (Note we did *not* need $\partial/\partial p$ of the normalization (15); the cancellation is purely a consequence of the left eigen-relation. Differentiating (15) gives the complementary fact $2\,\mathrm{Re}(\boldsymbol{\phi}_n^{\mathsf{H}}\mathbf{M}_r\boldsymbol{\phi}_n') = -\boldsymbol{\phi}_n^{\mathsf{H}}\mathbf{M}_r'\boldsymbol{\phi}_n$, which fixes the component of $\boldsymbol{\phi}_n'$ along $\boldsymbol{\phi}_n$ but is not needed here.)

Result. What remains of (17) is the Hellmann–Feynman sensitivity

$$\boxed{\frac{\partial\lambda_n}{\partial p} = \boldsymbol{\phi}_n^{\mathsf{H}}\Big(\frac{\partial\mathbf{K}_r}{\partial p} - \lambda_n\frac{\partial\mathbf{M}_r}{\partial p}\Big)\boldsymbol{\phi}_n} \tag{5}$$

and then $\partial\omega_n/\partial p = \partial\lambda_n/\partial p/(2\omega_n)$. The formula has a clean physical reading: $\boldsymbol{\phi}_n^{\mathsf{H}}\mathbf{K}_r'\boldsymbol{\phi}_n$ is the change in the mode's *elastic (strain) energy* caused by the design perturbation, and $\lambda_n\,\boldsymbol{\phi}_n^{\mathsf{H}}\mathbf{M}_r'\boldsymbol{\phi}_n$ the corresponding change in its *kinetic energy*; the eigenfrequency rises when the perturbation stiffens the mode faster than it mass-loads it, and falls otherwise. Everything on the right is available from the one eigenpair $(\lambda_n, \boldsymbol{\phi}_n)$ already computed — no extra solve, no eigenvector derivative.

Reduction to element energies. The reduced matrices are congruence transforms of the full-cell matrices, $\mathbf{K}_r = \mathbf{T}^{\mathsf{H}}\mathbf{K}\mathbf{T}$ and $\mathbf{M}_r = \mathbf{T}^{\mathsf{H}}\mathbf{M}\mathbf{T}$, with the Bloch reduction $\mathbf{T}(\mathbf{k})$ *independent* of the design. Hence $\mathbf{K}'_r = \mathbf{T}^{\mathsf{H}}\mathbf{K}'\mathbf{T}$ and likewise for $\mathbf{M}'_r$, and (5) becomes a quadratic form in the full-space Bloch state $\boldsymbol{\psi}_n = \mathbf{T}\boldsymbol{\phi}_n$:

$$\frac{\partial \lambda_n}{\partial p} = \boldsymbol{\psi}_n^{\mathsf{H}} \frac{\partial \mathbf{K}}{\partial p} \boldsymbol{\psi}_n - \lambda_n \, \boldsymbol{\psi}_n^{\mathsf{H}} \frac{\partial \mathbf{M}}{\partial p} \boldsymbol{\psi}_n. \tag{19}$$

Because $\mathbf{K} = \sum_e \mathbf{k}_e$ and $\mathbf{M} = \sum_e \mathbf{m}_e$ are element-assembled and each element depends on the design only locally, both quadratic forms collapse to sums of *element energies* of $\boldsymbol{\psi}_n$. For an orientation variable $\hat{\theta}_i$, only $\mathbf{K}$ depends on it (through $\mathbf{k}_f(\theta)$), and the dependence passes through the CS-RBF map $\theta = \hat{\boldsymbol{\Phi}}\hat{\boldsymbol{\theta}}$, so by the chain rule

$$\frac{\partial \lambda_n}{\partial \hat{\theta}_i} = \sum_e \hat{\Phi}_{ei} \, z_e^{\,p} \, \boldsymbol{\psi}_{n,e}^{\mathsf{H}} \frac{\partial \mathbf{k}_f}{\partial \theta}(\theta_e) \, \boldsymbol{\psi}_{n,e}, \tag{20}$$

with $\partial \mathbf{k}_f / \partial \theta$ the exact Fourier derivative (**??**). For a density variable z_e, the two-phase SIMP interpolation gives $\partial \mathbf{k}_e / \partial z_e = p \, z_e^{p-1} (\mathbf{k}_f - \mathbf{k}_m)$ and $\partial \mathbf{m}_e / \partial z_e = (\rho_f - \rho_m)\mathbf{m}_e^0$, so

$$\frac{\partial \lambda_n}{\partial z_e} = p \, z_e^{p-1} \, \boldsymbol{\psi}_{n,e}^{\mathsf{H}} (\mathbf{k}_f - \mathbf{k}_m) \boldsymbol{\psi}_{n,e} - \lambda_n (\rho_f - \rho_m) \, \boldsymbol{\psi}_{n,e}^{\mathsf{H}} \mathbf{m}_e^0 \boldsymbol{\psi}_{n,e}. \tag{21}$$

These per-band, per-$\mathbf{k}$ gradients are then combined by the smooth KS extrema of (6) (`flatband_opt.py`) and finite-difference verified to $\sim 10^{-6}$.

Caveat: degenerate eigenvalues. The cancellation above assumed λ_n simple, so that $\boldsymbol{\phi}_n$ is defined up to a phase and (B)= λ_n·(D) holds for *that* eigenvector. At a degenerate eigenvalue — e.g. the four-fold Γ-point degeneracy of a symmetric cell — the eigenvectors within the degenerate subspace are only defined up to a unitary mixing, the map $p \mapsto \lambda_n(p)$ is generally not differentiable (only directionally so), and (5) evaluated on an arbitrary basis of the subspace disagrees with finite differences by order one. This is not a defect of the formula but of the question: one must instead diagonalize the 2×2 (or $d \times d$) restriction of $\mathbf{K}'_r - \lambda_n \mathbf{M}'_r$ onto the degenerate subspace to get the (one-sided) sensitivities of the split branches. In practice we sidestep this by starting the optimizer from a deliberately symmetry-broken cell, where all bands of interest are simple; §3 records the failure mode when this is neglected.

.4 Eigenvalue sensitivity for the finite bracket

The bracket (§5) maximizes the fundamental eigenvalue $\lambda_1 = \omega_1^2$ of the clamped generalized problem $\mathbf{K}(z, \theta) \, \boldsymbol{\phi} = \lambda \, \mathbf{M}(z) \, \boldsymbol{\phi}$, with $\boldsymbol{\phi}$ mass-normalized, $\boldsymbol{\phi}^{\mathsf{T}} \mathbf{M} \boldsymbol{\phi} = 1$. For a simple eigenvalue the Rayleigh-quotient derivative w.r.t. any design parameter p is

$$\frac{\partial \lambda_1}{\partial p} = \boldsymbol{\phi}^{\mathsf{T}} \Big(\frac{\partial \mathbf{K}}{\partial p} - \lambda_1 \frac{\partial \mathbf{M}}{\partial p} \Big) \boldsymbol{\phi}, \tag{22}$$

the eigenvector's own variation dropping out at first order. Orientation leaves the mass untouched ($\partial \mathbf{M} / \partial \theta = 0$), so with the five-term decomposition (**??**), $\partial \lambda_1 / \partial \theta_e = w(z_e) \, \boldsymbol{\phi}_e^{\mathsf{T}} (\partial \mathbf{k}_f / \partial \theta_e) \boldsymbol{\phi}_e$. Density moves both operators, but *differently*: the stiffness is SIMP-penalized, $\mathbf{k}_e = w(z_e)\mathbf{k}_f + (1 - w(z_e))\mathbf{k}_m$ with $w(z) = z^p$, while the element mass is *linear* in the density, $\mathbf{m}_e = [z_e \rho_f + (1 -$

$z_e)\rho_m]\,\mathbf{m}_0$ — the standard modified-SIMP mass interpolation that keeps low-density regions from carrying spurious low-frequency modes. Hence

$$\frac{\partial \lambda_1}{\partial z_e} = \underbrace{w'(z_e)\,\boldsymbol{\phi}_e^{\mathsf{T}}(\mathbf{k}_f - \mathbf{k}_m)\boldsymbol{\phi}_e}_{\text{stiffen where the mode strains}} - \underbrace{\lambda_1(\rho_f - \rho_m)\,\boldsymbol{\phi}_e^{\mathsf{T}}\mathbf{m}_0\boldsymbol{\phi}_e}_{\text{remove mass at the antinodes}}. \tag{23}$$

The second, negative term is the counterintuitive one and is the whole point of the example: $\boldsymbol{\phi}_e^{\mathsf{T}}\mathbf{m}_0\boldsymbol{\phi}_e$ is the element's kinetic energy in the mode, so *removing* material from the mode's antinodes *raises* λ_1. At a fixed mass budget the optimizer therefore moves material toward the strained regions and away from the antinodes, lifting ω_1 far more than a uniform stiffening would. Both (22)–(23) pass the finite-difference check to a gradient-normalized $\lesssim 10^{-6}$, and the θ part isotropy-annihilates ($\partial\lambda_1/\partial\theta \to 0$ as $\mathbf{Q}_f \to$ isotropic, Remark **??**) to $O(10^{-18})$.

.5 p-norm stress sensitivity

The bracket's stress objective is the peak von Mises stress under the tip load, smoothed by the p-norm surrogate

$$\Sigma_p = \Big(\sum_e \tfrac{A_e}{\sum_k A_k}\,\tilde{\sigma}_e^{\,p}\Big)^{1/p}, \qquad \tilde{\sigma}_e = z_e^{\,q}\sqrt{\boldsymbol{\sigma}_e^{\mathsf{T}}\mathbf{V}\boldsymbol{\sigma}_e}, \quad \boldsymbol{\sigma}_e = \mathbf{D}_f(\theta_e)\,\mathbf{B}_e\mathbf{u}_e, \tag{24}$$

where $\mathbf{u}$ solves the static problem $\mathbf{K}(z,\theta)\mathbf{u} = \mathbf{F}_s$, $\mathbf{V}$ is the plane-stress von Mises matrix (so $\boldsymbol{\sigma}^{\mathsf{T}}\mathbf{V}\boldsymbol{\sigma} = \sigma_{xx}^2 - \sigma_{xx}\sigma_{yy} + \sigma_{yy}^2 + 3\tau_{xy}^2$), and $z_e^{\,q}$ ($q = \frac{1}{2}$) is a qp-relaxation that stops void elements from carrying spurious stress. The design derivative has an explicit part — the element stress depends on z through the qp factor and on θ through $\mathbf{D}_f$ — and an implicit part through $\mathbf{u}$; the latter is a single adjoint solve reusing the static factorization, $\mathbf{K}\boldsymbol{\mu} = \partial\Sigma_p/\partial\mathbf{u}$, after which $\partial\Sigma_p/\partial p = \partial\Sigma_p/\partial p|_{\text{expl}} - \boldsymbol{\mu}^{\mathsf{T}}(\partial\mathbf{K}/\partial p)\,\mathbf{u}$ for $p \in \{z,\theta\}$ ($\mathbf{K}$ symmetric). The z^{q-1} in the explicit qp term is regular once carried by the p-norm weight (whose factor $\tilde{\sigma}_e^{\,p-1} \to 0$ as $z_e \to 0$); both $\partial\Sigma_p/\partial z$ and $\partial\Sigma_p/\partial\theta$ pass the finite-difference check to a gradient-normalized $\lesssim 10^{-7}$.

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